

RA8P1 E-READER

Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.

128 MiB Octal NOR

File: nor_flash.kicad_sch

RABP1 IO allocation

SDRAM_CS_N
SDRAM_RAS_N
SDRAM_WE_N
SDRAM_CKE
SDRAM_CAS_N
SDRAM_A[0..12]
SDRAM_DQ[0..31]
SDRAM_CLK
SDRAM_DQM[0..3]
SDRAM_BA[0..1]

File: mcu_interfaces.kicad_sch

RABP1 internal core regulator

File: mcu_core_power.kicad_sch

RABP1 IO and analog supplies

File: mcu_supply_power.kicad_sch

RABP1 clocks, reset and debug

File: mcu_clocks_debug.kicad_sch

Global supplies: +3V3_MCU and GND.
MCU_VCORE remains local to the processor core-power sheet.
Circuit notes link to the supporting engineering calculations.
Footprint and PCB qualification are deferred by scope.

Page and volume buttons

PAGE_PREV_N
PAGE_NEXT_N
VOL_DOWN_N
VOL_UP_N

File: user_controls.kicad_sch

ESP32-C6 radio

RADIO_CS_N
RADIO_COPI
RADIO_HANDSHAKE_N
RADIO_DATA_READY_N
RADIO_SCLK
RADIO_CIPO
RADIO_RESET_REQ_N
RADIO_PWR_EN
MCU_RESET_N

File: radio_esp32.kicad_sch

64 MiB SDRAM

SDRAM_DQM[0..3]
SDRAM_A[0..12]
SDRAM_WE_N
SDRAM_DQ[0..31]
SDRAM_CS_N
SDRAM_CKE
SDRAM_CAS_N
SDRAM_RAS_N
SDRAM_BA[0..1]
SDRAM_CLK

File: sdram.kicad_sch

Power button and source control

POWER_BUTTON_N
POWER_KILL_N

File: power_button.kicad_sch

Main 3V3 digital supply

File: main_3v3_supply.kicad_sch

MAIN_PWR_EN

Sheet: /
File: ereader_rev1.kicad_sch

Title: RA8P1 e-reader – schematic index

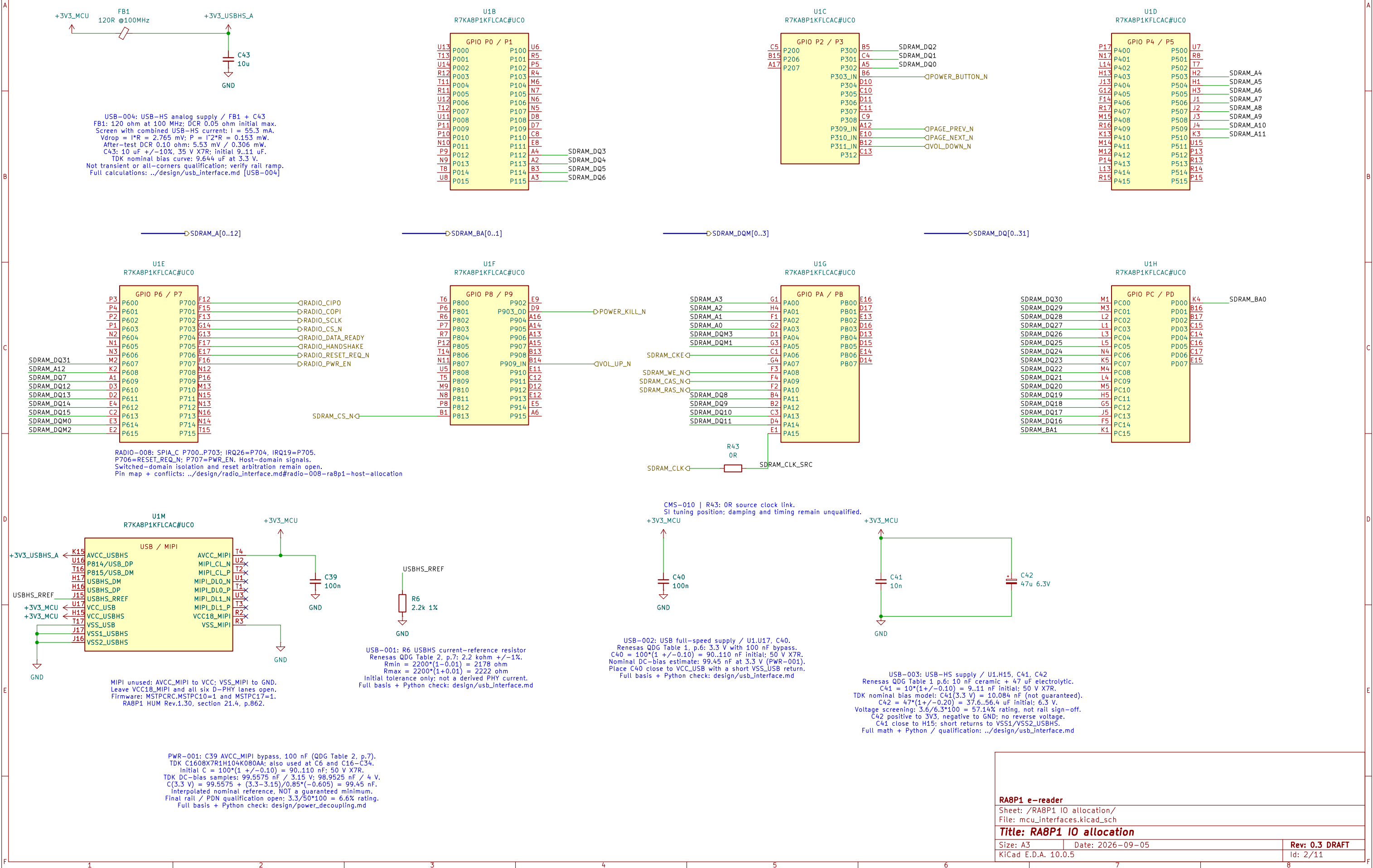
Size: A3 Date: 2026-09-05

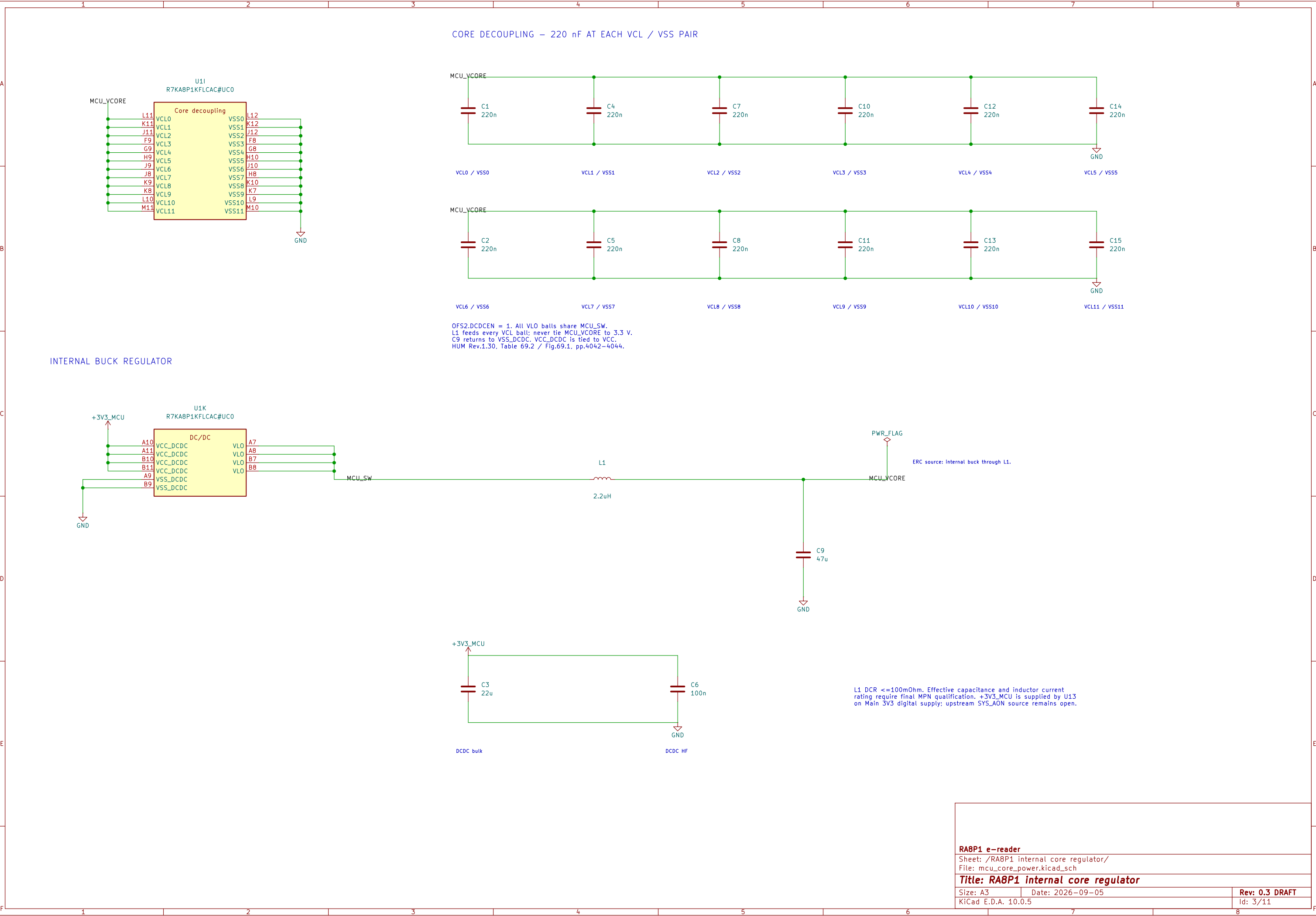
KiCad E.D.A. 10.0.5

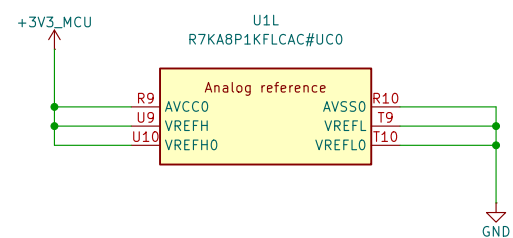
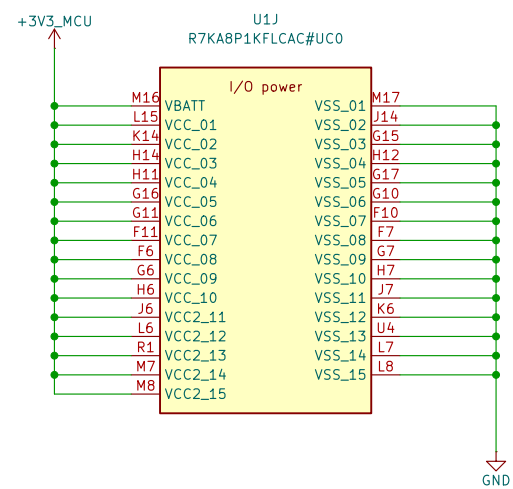
Rev: 0.3 DRAFT

Id: 1/11

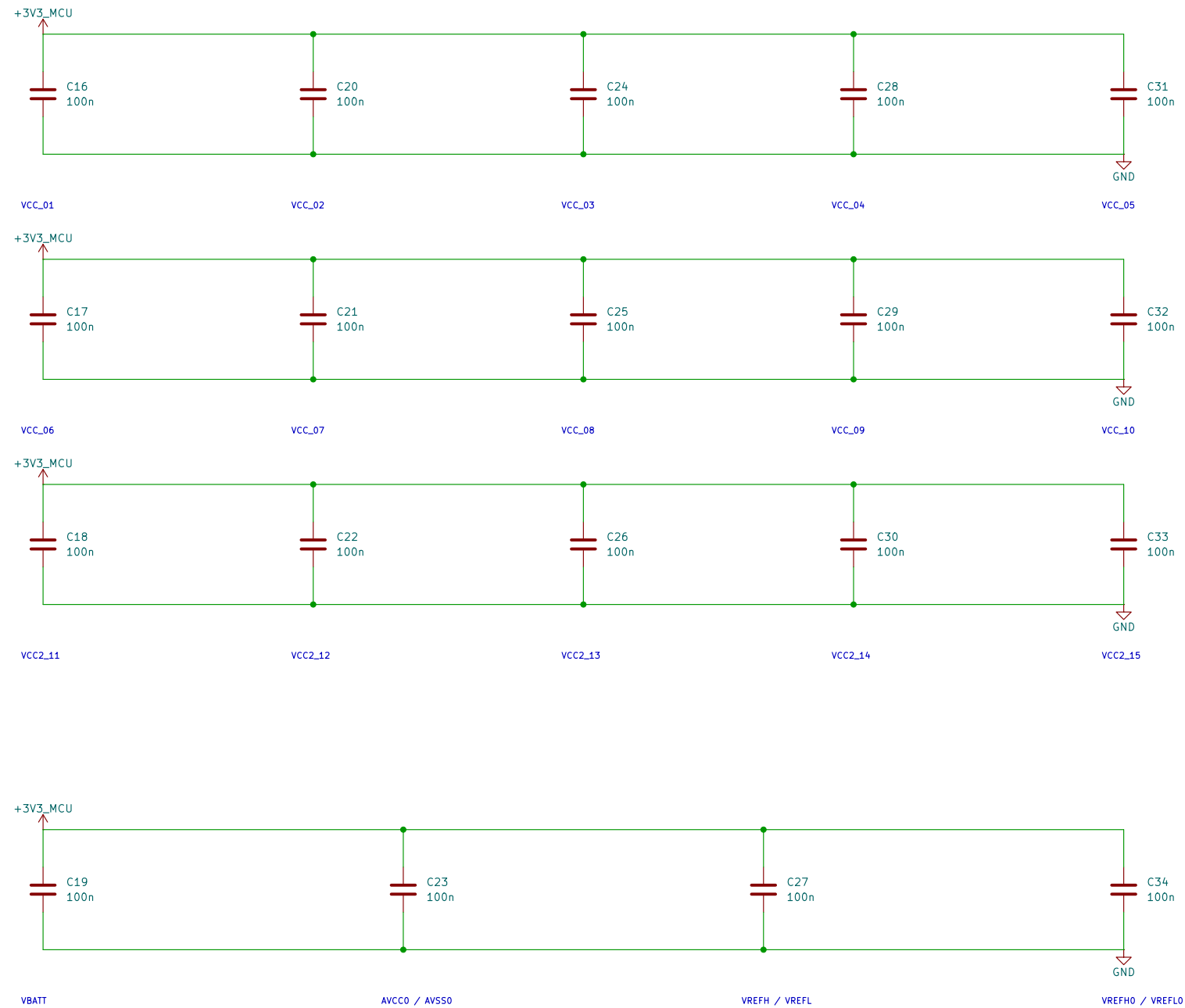
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



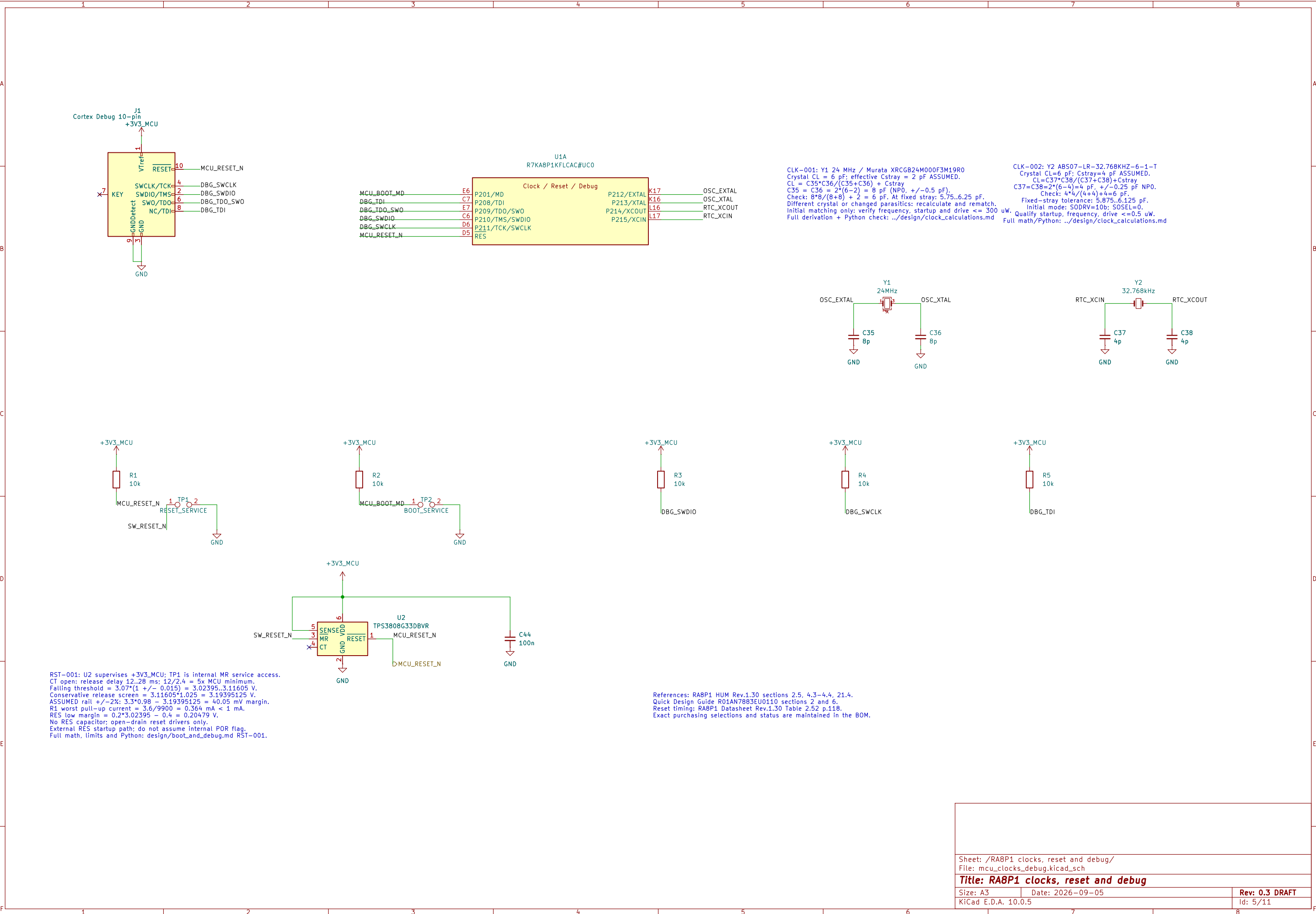




I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR



VCC and VCC2 use 3.3 V. All returns use one board ground plane. VBATT follows the main rail; no separate backup-cell supply. Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs. USB/MIP1 supplies remain on the interface sheet and are not yet wired. HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861-862. This is an electrical draft; capacitor MPN qualification remains open.

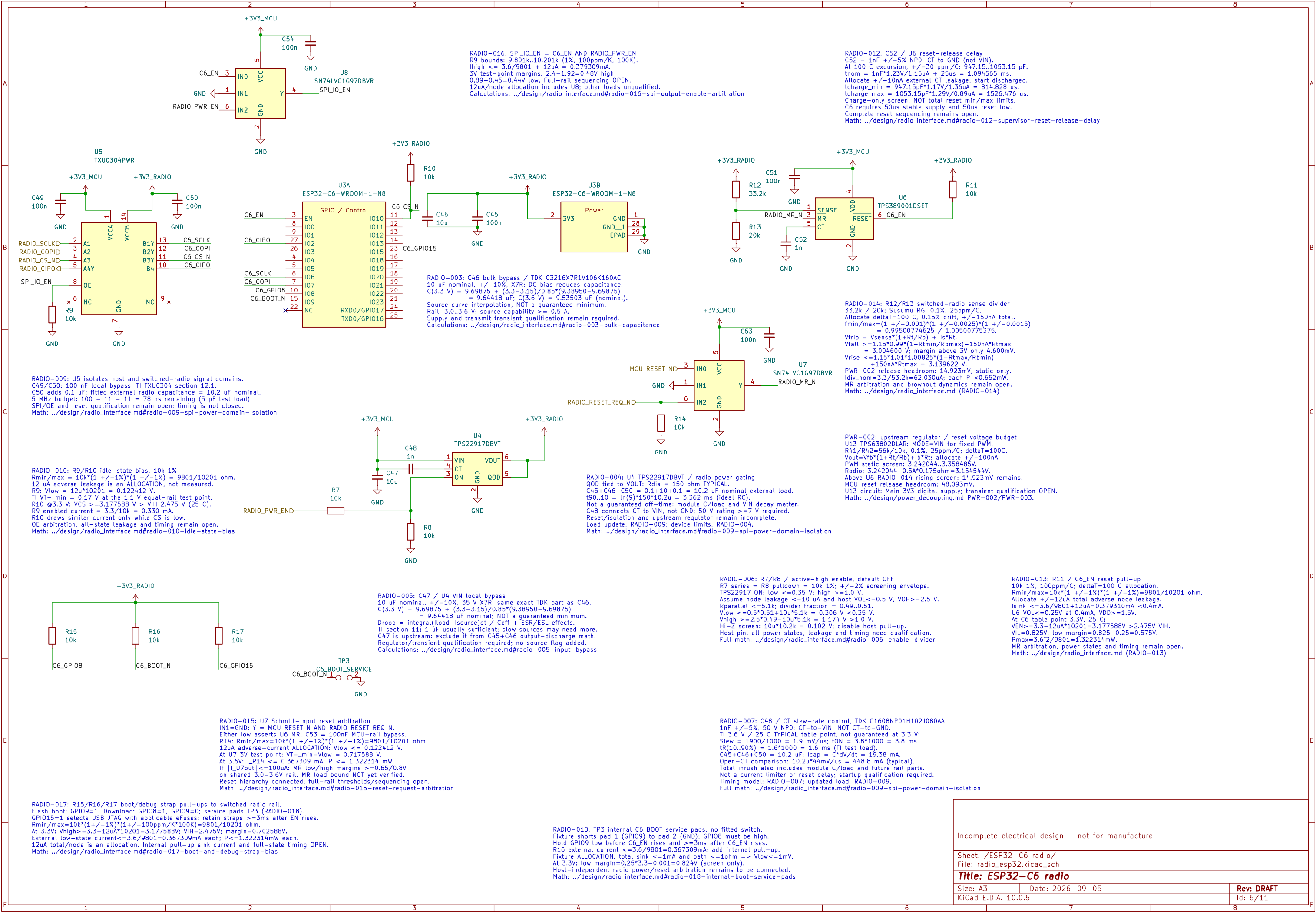


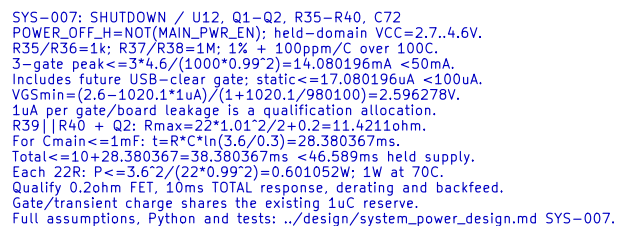
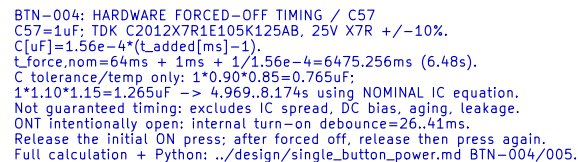
CLK-001: Y1 24 MHz / Murata XRCGB24M000F3M19R0
Crystal CL = 6 pF; effective Cstray = 2 pF ASSUMED.
CL = C35*C36/(C35+C36) + Cstray
C35 = C36 = 2*(6-2) = 8 pF (NP0, +/-0.5 pF).
Check: 8*8/(8+8) + 2 = 6 pF. At fixed stray: 5.75..6.25 pF.
Different crystal or changed parasitics: recalculate and rematch.
Initial matching only: verify frequency, startup and drive <= 300 uW.
Full derivation + Python check: ../design/clock_calculations.md

CLK-002: Y2 ABS07-LR-32.768KHZ-6-1-T
Crystal CL=6 pF; Cstray=4 pF ASSUMED.
CL=C37*C38/(C37+C38)+Cstray
C37=C38=2*(6-4)=4 pF, +/-0.25 pF NP0.
Check: 4*4/(4+4)+4=6 pF.
Fixed-stray tolerance: 5.875..6.125 pF.
Initial mode: SODRV=10b; SOSEL=0.
Qualify startup, frequency, drive <=0.5 uW.
Full math/Python: ../design/clock_calculations.md

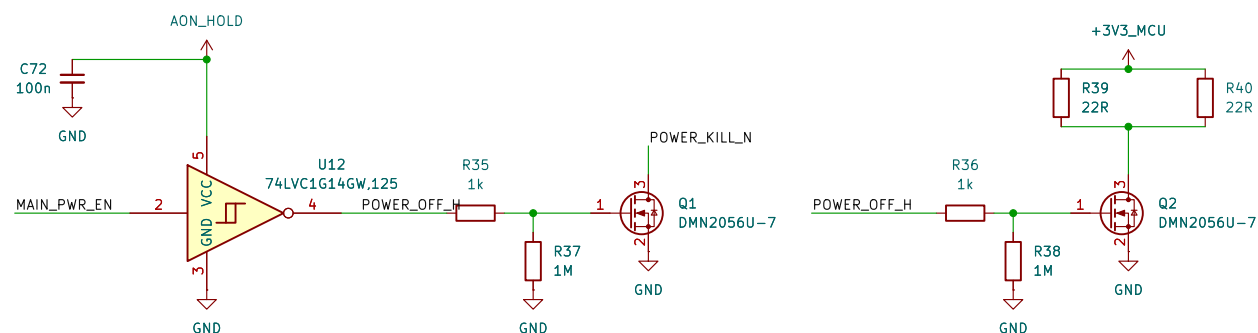
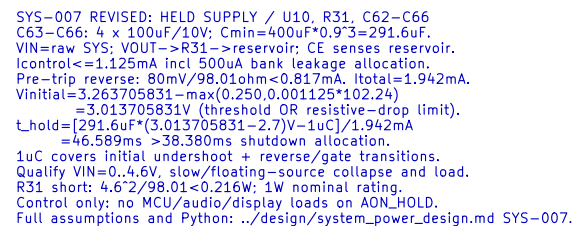
RST-001: U2 supervises +3V3_MCU; TP1 is internal MR service access.
CT open: release delay 12..28 ms; 12/2.4 = 5x MCU minimum.
Falling threshold = 3.07*(1 +/- 0.015) = 3.02395..3.11605 V.
Conservative release screen = 3.11605*1.025 = 3.19395125 V.
ASSUMED rail +/-2%: 3.3*0.98 - 3.19395125 = 40.05 mV margin.
R1 worst pull-up current = 3.6/9900 = 0.364 mA < 1 mA.
RES low margin = 0.2*3.02395 - 0.4 = 0.20479 V.
No RES capacitor; open-drain reset drivers only.
External RES startup path; do not assume internal POR flag.
Full math, limits and Python: design/boot_and_debug.md RST-001.

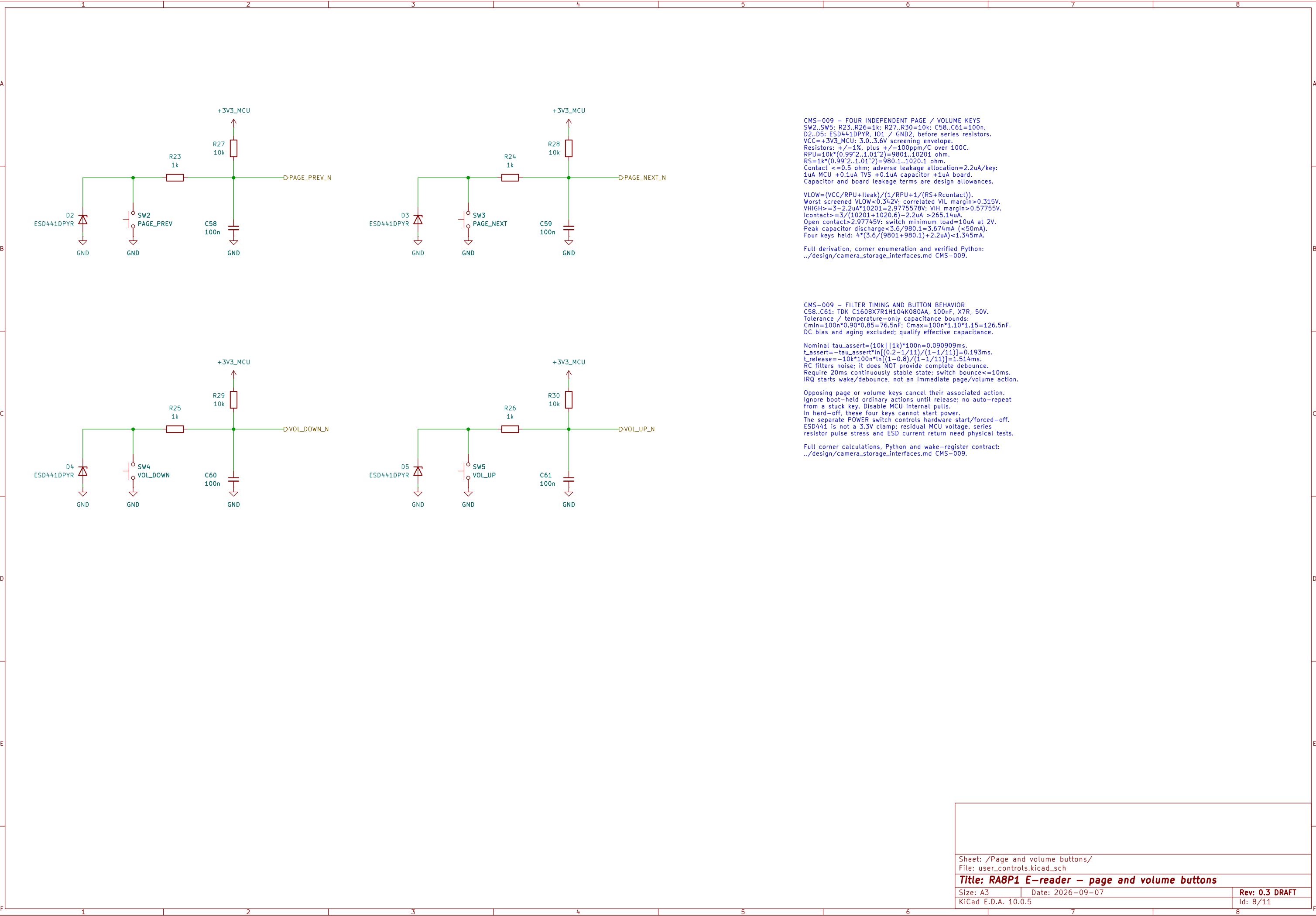
References: RABP1 HUM Rev.1.30 sections 2.5, 4.3-4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RABP1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.





ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.





CMS-009 – FOUR INDEPENDENT PAGE / VOLUME KEYS
SW2..SW5; R23..R26=1k; R27..R30=10k; C58..C61=100n.
D2..D5: ESD441DPYR, IO1 / GND2, before series resistors.
VCC=+3V3_MCU; 3.0..3.6V screening envelope.
Resistors: +/-1%, plus +/-100ppm/C over 100C.
RPU=10k*(0.99^2..1.01^2)=9801..10201 ohm.
RS=1k*(0.99^2..1.01^2)=9801..10201 ohm.
Contact <=0.5 ohm; adverse leakage allocation=2.2uA/key:
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

VLOW=(VCC/RPU+Ileak)/(1/RPU+1/(RS+Rcontact)).
Worst screened VLOW<0.342V; correlated VIL margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; VIH margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact>2.97745V; switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/980.1=3.674mA (<50mA).
Four keys held: 4*(3.6/(9801+980.1))+2.2uA<1.345mA.

Full derivation, corner enumeration and verified Python:
../design/camera_storage_interfaces.md CMS-009.

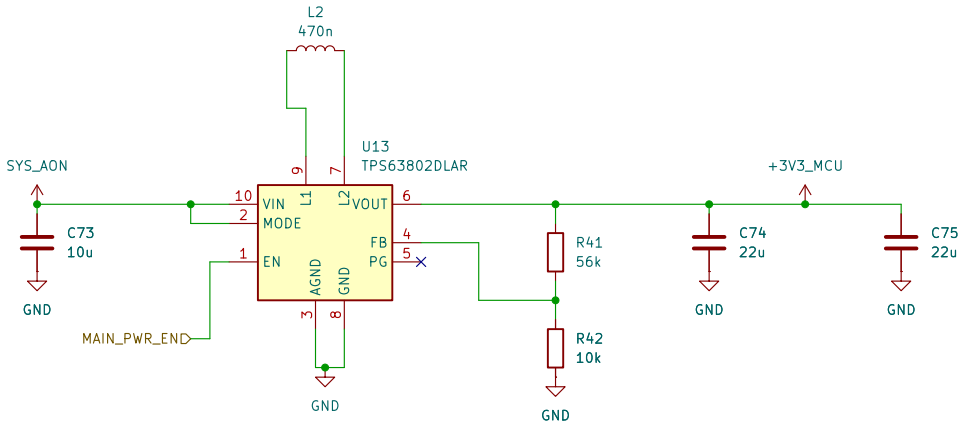
CMS-009 – FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
Cmin=100n*0.90*0.85=76.5nF; Cmax=100n*1.10*1.15=126.5nF.
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k||1k)*100n=0.090909ms.
t_assert=-tau_assert*ln[(0.2-1/11)/(1-1/11)]=0.193ms.
t_release=-10k*100n*ln[(1-0.8)/(1-1/11)]=1.514ms.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce<=10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot-held ordinary actions until release; no auto-repeat
from a stuck key. Disable MCU internal pulls.
In hard-off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced-off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
../design/camera_storage_interfaces.md CMS-009.

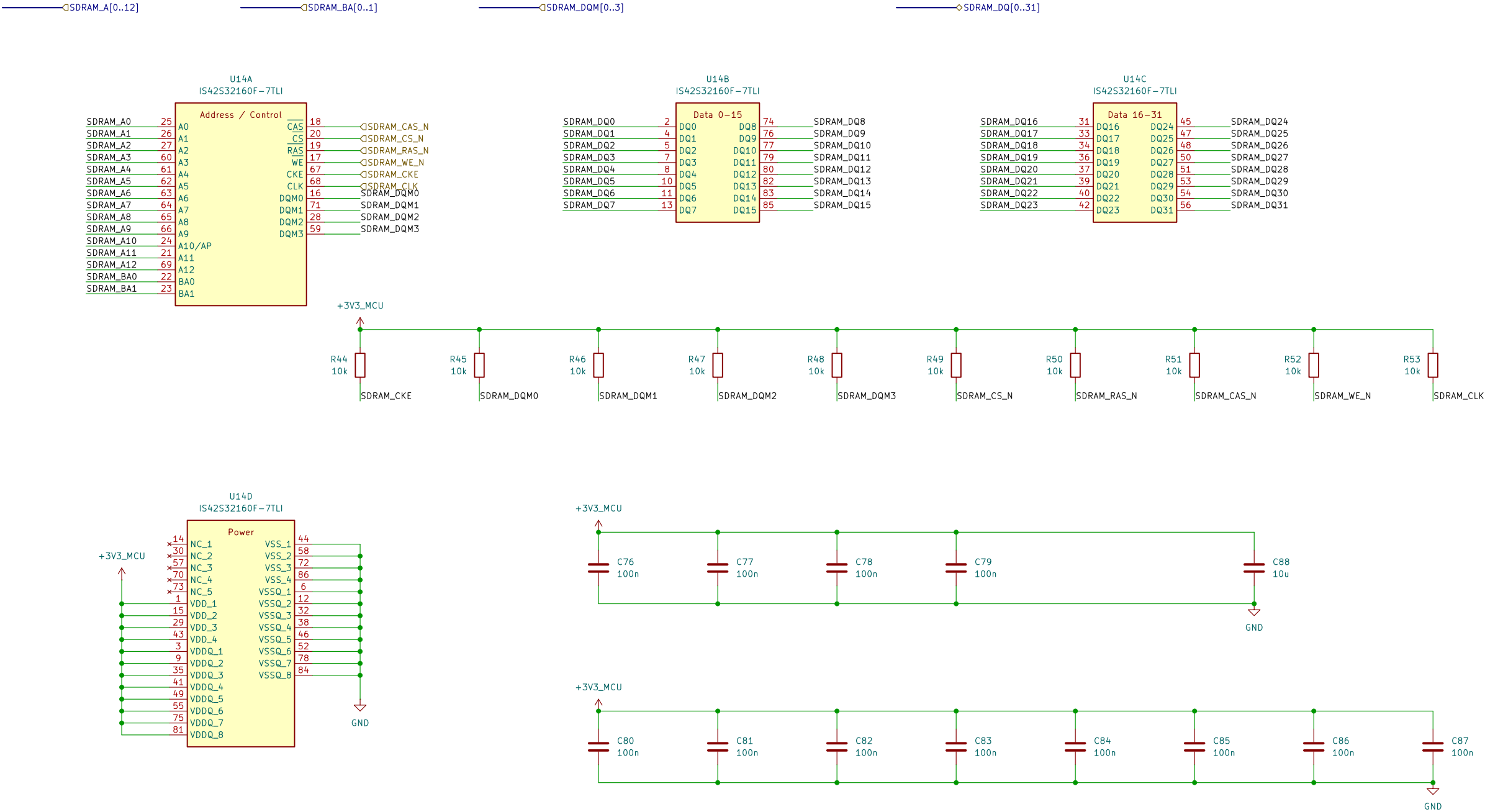
MAIN DIGITAL 3.3 V SUPPLY
TPS63802 buck–boost; hardware–controlled enable; fixed PWM.
Engineering draft: upstream source and corner qualification remain open.



PWR–002 / PWR–003: SETPOINT AND LOCAL COMPONENTS
U13 TPS63802DLAR; MODE=VIN, fixed PWM.
R41=56k; R42=10k; each +/-0.1%, 25ppm/C.
Vnom=0.500*(1+56k/10k)=3.300V.
Idivider=3.300/(56k+10k)=50uA.
For 100C resistor–temperature change:
fLo=0.999*0.9975=0.9965025; fHi=1.001*1.0025=1.0035025.
Vmin=0.495*(1+56k*fLo/(10k*fHi))–100nA*56k*fHi
=3.242044V.
Vmax=0.505*(1+56k*fHi/(10k*fLo))+100nA*56k*fHi
=3.358485V.
Static screen only; ripple, load steps and drift still require validation.
C73: CIN_model=10u*0.90*0.85*0.60=4.590uF >=4uF.
C74+C75: COUT_model=2*22u*0.90*0.85*0.60=20.196uF >=7uF.
0.90=tolerance; 0.85=X7R temperature; 0.60=residual ALLOCATION.
Qualify bias, aging, excitation and assembly; 60% is not guaranteed.
L2: 0.47uH +/-20% gives 0.376..0.564uH INITIAL only.
Required installed effective L=0.37..0.57uH; bias/reflow/hot tests OPEN.
TI SLVSEU9D Table 7–1, sections 9.4/10.2; exact parts in native BOM.
Full derivation, primary sources and executable Python:
../design/power_decoupling.md PWR–002/PWR–003.

PWR–003: LOAD, SOURCE AND STARTUP ACCEPTANCE LIMITS
Main continuous allocation: 0.750 MCU + 0.500 radio
+ 0.250 SDRAM + 0.250 NOR + 0.050 logic = 1.800A.
These are qualification allocations, not measured maximum currents.
SD, camera, audio, display/touch/frontlight need separate SYS supplies.
Pout=3.358485094*1.80=6.045273W.
VIN>=3.20V allocation; raw trip min=3.263705831V.
Allowed raw–to–VIN loss=3.263705831–3.20=63.705831mV.
At ASSUMED eta=85%: Iin=Pout/(eta*3.20)=2.222527A.
Ploss=Pout*(1/eta–1)=1.066813W.
Reference rise=Ploss*81C/W=86.411846C; NOT enclosure prediction.
Verify hot current capability and junction temperature <125C with margin.
Cold subtotal, RADIO OFF: 1.330 DCDC + 0.550 other = 1.880A.
Radio on would give 2.380A > nominal 2A; keep radio OFF through start/wake.
Largest wake–reference subtotal=1.270+0.550=1.820A.
DCDC figures are reference–only; non–DCDC start and extra charging excluded.
Additional charging: I=Cadditional*dV/dt; qualify the actual ramp.
Total directly connected main–rail C <=1mF for SYS–007 discharge budget.
No externally driven +3V3_MCU: forced PWM can sink reverse current.
PWR–002 static release margin: radio 14.922537mV; MCU 48.092861mV.
Ripple, startup and transients must fit; no fabrication release.
Full sources, derivation and Python: ../design/power_decoupling.md PWR–003;
source–loss shutdown: ../design/system_power_design.md SYS–007.

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

Sheet: /64 MiB SDRAM/
File: sdram.kicad_sch

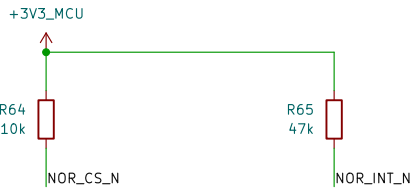
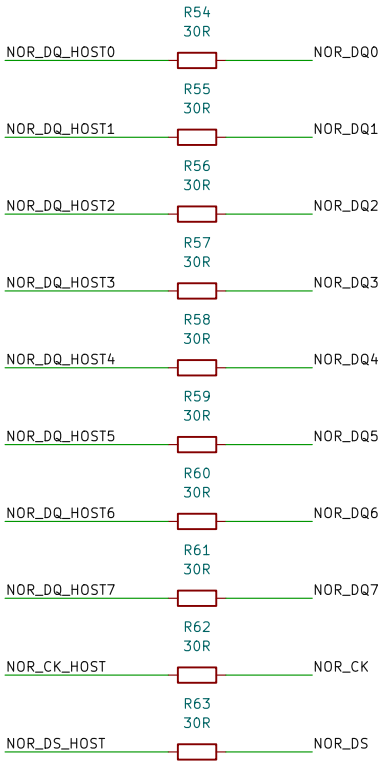
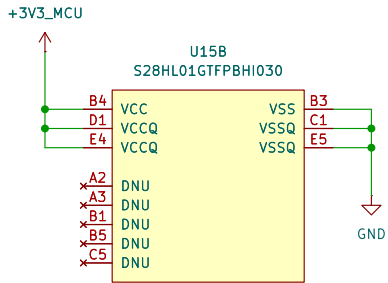
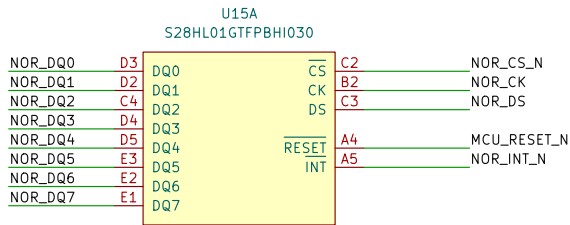
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Size: A3 Date: 2026-09-08

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Rev: 0.3 DRAFT

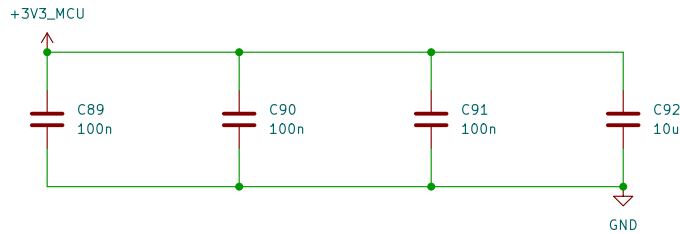
Id: 10/11



CMS-011 | NOR series and idle states
R54-R63: 10 x 30R, 0.1%, 25 ppm/C.
100 C departure from reference temperature:
 $R_{min} = 30 \times 0.999 \times 0.9975 = 29.895075 \text{ ohm}$
 $R_{max} = 30 \times 1.001 \times 1.0025 = 30.105075 \text{ ohm}$
Within JEDEC251 30R +/-1%; 29.7..30.3 ohm.
External parts remain required; SI check OPEN.

R64 CS pull: 10k; R65 INT pull: 47k.
1%, 100 ppm/C, 100 C: factors 0.9801..1.0201.
At 4uA adverse leakage, rail min 3.242044 V:
 $CS_{Vhigh} \geq 3.242044 - 4u \times 10201 = 3.201240 \text{ V}$
 $INT_{Vhigh} \geq 3.242044 - 4u \times 47944.7 = 3.050265 \text{ V}$
 $INT_{Ilow} \leq 3.6/46064.7 + 2u = 80.150949 \text{ uA}$
< 100uA flash VOL test: 47k departs from 5-10k.
RC timing and board leakage qualification OPEN.

Linked CMS-011B/C/F: sources and Python checks.
Host hierarchy and shared reset hookup pending.



CMS-011 | NOR supply and bypass
U15B VCC and VCCQ share +3V3_MCU (2.7 to 3.6 V).
C89-C91: 3 x 100 nF, one per supply ball.
C92: 10 uF local bulk.
 $C_{nom} = 3 \times 0.1 + 10 = 10.3 \text{ uF}$
Initial +/-10%; $C_{min}/max = 10.3 \times (0.9 / 1.1)$
 $= 9.27 / 11.33 \text{ uF}$: not an effective-C guarantee.
Included in the PWR-003 1 mF whole-rail limit.
DC bias, temperature, aging and PDN qualification OPEN.
Linked CMS-011B/F: sources and Python-verified arithmetic.

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KiCad E.D.A. 10.0.5

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